A Universal Geometric Theory of Nuclear Reactions in CPTG

Post-Silicon Reduced-Graph Reachability and the Gap-Free $A = 1$–119 Mass-Sector Register

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Abstract

This computational companion documents an exploratory extension of the Curvature Polarization Transport Gravity (CPTG) nuclear-reaction program beyond the native reaction-network inventory at silicon-30. Combining the native $A = 1$–30 inventory, an explicitly labeled mass-five resonance-gap row, and a prescribed-trajectory graph containing forward neutron-capture and beta-minus edges yields an archived, gap-free mass-sector register through $A = 119$. An independent replay reproduces the accepted $A \leq 120$, eight-substep result to a maximum relative difference of 1.75×10^{-15}. Because the reduced graph is one-way and acyclic, moving the absorbing boundary to $A = 140$ and $A = 160$ is expected to leave upstream $A \leq 119$ values unchanged; the observed identity is therefore an implementation and indexing check rather than an independent physical validation. Temporal refinement through 64 substeps preserves positive support while changing selected $A = 31$–119 checkpoints by at most approximately 1.06% from 32 to 64 substeps. Exact zero-source and source-scaling controls confirm source isolation and linearity under the fixed background trajectories. Removing early silicon-30 source values below 10^{-50} preserves positive graph support in the enlarged qualification graph but strongly suppresses deep-tail magnitudes. The result therefore establishes reduced-graph reachability under the stated numerical construction, not precision-qualified primordial heavy-element abundances, a finite physical endpoint, or an independent test of gravitational-to-nuclear structural continuity.

1 Introduction

This paper is the computational companion to *A Universal Geometric Theory of Nuclear Reactions in CPTG: The Four-Sector Foundation of Vertex, Bridge, Closure, and Saturation* [1]. It supplies the detailed numerical authority for the exploratory post-silicon reachability result summarized there. The Curvature Polarization Transport Gravity (CPTG) nuclear-reaction program begins with the free-nucleon vertex, proceeds through the deuterium bridge and the mass-three closure pair, and reaches helium-four saturation under the source, amplitude, rate, transport, and abundance-closure structure developed in the universal-theory paper. That broader theory originated in the commissioned CPTG nuclear-reaction study [2]. The present manuscript follows the represented native reaction-network inventory through oxygen-16 and silicon-30, then documents a prescribed-trajectory neutron-capture and beta-minus continuation of the nuclide graph.

The detailed coefficient construction of the universal four-sector law can be found in the companion theory paper and is not repeated here. For masses above $A = 30$, this addendum begins from the stored native thermodynamic, neutron, and silicon-30 trajectories and applies selected JINA ReacLib neutron-capture and beta-minus edge rates through `pynucastro` [5, 6]. The post-silicon calculation is an output-layer reachability and numerical-qualification study seeded by the native CPTG result.

The purpose of this extension is to identify, organize, and document a complete bookkeeping register for the mass sectors and graph states used by the combined construction. Every integer mass number in the interval

$$1 \leq A \leq 119 \tag{1}$$

appears exactly once in Table 5. Mass five is included as an explicitly labeled physical-gap annotation through the unbound helium-five and lithium-five resonances relevant to the $n + {}^{4}\mathrm{He}$ and $p + {}^{4}\mathrm{He}$ systems; neither resonance is an abundance-bearing network coordinate or a post-silicon graph node [7].

The register contains four distinct support categories. The abundance-bearing entries at $A = 1$–4 and $A = 6$–30 come from the native reaction-network inventory. The $A = 5$ row is the resonance-gap annotation described above and is not a native abundance coordinate. Masses $A = 31$–32 form the first post-silicon prescribed-trajectory block and have a separate higher-resolution temporal diagnostic. Masses $A = 33$–119 belong to the larger exploratory neutron-capture and beta-minus graph. These categories preserve the distinction between native-network coverage, the mass-five annotation, the separately refined first post-silicon block, and the deeper exploratory continuation.

The mass-120 sector in the archived calculation is an absorbing computational boundary and is excluded from physical interpretation. The qualification moves that boundary outward to $A = 140$ and $A = 160$. In the present one-way acyclic graph, upstream boundary invariance is guaranteed by topology; the outward-boundary runs therefore test implementation consistency and the absence of unintended back-coupling. Temporal refinement and source-cutoff controls, rather than the boundary shift itself, determine the numerical sensitivity of the published register.

The scientific purpose of this addendum is narrower than that of the companion theory paper. It records the post-silicon computational extension, its mass-sector register, and the controls that determine which parts of the continuation are numerically stable. The broader theoretical interpretation is summarized only to identify the origin of the structure being extended.

2 Relation to the CPTG Framework

The comparison parameters carried into the universal nuclear-reaction theory were introduced in the published *CPTG Geometric Pi Branch: Comparison Coordinates* paper [3]. Curvature polarization and curvature transport provide the broader CPTG interpretation of structural response and directed propagation. In the nuclear construction, that distinction is reflected in differentiated reaction-source states and baryon-conserving transport through the ordered light-nuclide sequence.

Structural quantity	CPTG role	Role in the nuclear calculation	Status
Curvature polarization	Structural response to baryonic organization	Differentiates the reaction-source sectors and their constrained polarization modes	Inherited; not refitted
Curvature transport	Directed propagation of organized curvature	Organizes baryon-conserving transfer along the ordered light-nuclide path	Inherited; not refitted
Native trajectory	Completed nuclear-network output	Supplies fixed T_9, ρ_b, Y_n, and $Y_{^{30}\mathrm{Si}}$ histories	Fixed input
Reaction data	Reaction-specific physical input	Supplies the selected ReacLib (n, γ) and β^- edge rates	Fixed rate-library input

Table 1: Structural inheritance and fixed inputs of the post-silicon continuation. No quantity in the table is adjusted to the post-silicon output.

Table 1 separates inherited CPTG structure from fixed trajectory and reaction inputs. The present calculation is conditioned on that inherited structure but does not independently test whether a gravitational-to-nuclear transfer is valid. Its narrower task is to document the exploratory continuation seeded by the native silicon-30 result, the gap-free mass-sector register, the mass-five interpretation, and the replay, temporal, source, and boundary implementation controls. The companion universal-theory paper remains the authority for the proposed nuclear law and any broader cross-scale interpretation.

3 Evidence Chain and Prescribed-Trajectory Continuation

The universal light-state sequence is

$$\{n, p\} \longrightarrow {}^{2}\mathrm{H} \longrightarrow \{{}^{3}\mathrm{H}, {}^{3}\mathrm{He}\} \longrightarrow {}^{4}\mathrm{He}. \tag{2}$$

For every abundance-bearing nuclide i, Y_i denotes the standard molar abundance used by the reaction-network convention; throughout this paper it is reported as the corresponding number abundance per baryon. The mass-sector abundance used throughout the continuation and register is the sum over all represented isobars at fixed mass number,

$$Y_A = \sum_{i:A_i=A} Y_i. \tag{3}$$

Thus Y_A is a number-abundance sum, not the mass fraction AY_A.

The sequence in Equation (2) does not terminate at the absence of a persistent mass-five state. The represented network bypasses the mass-five gap and reaches mass six and higher through multi-body and capture channels. The native basis proceeds through oxygen-16. An extended native inventory reaches silicon-30 only after addition of the missing bridge below, which activates the already represented downstream routes:

$${}^{22}\mathrm{Ne}(n,\gamma){}^{23}\mathrm{Ne} \xrightarrow{\beta^-} {}^{23}\mathrm{Na} + e^- + \bar{\nu}_e. \tag{4}$$

The first post-silicon continuation is

$${}^{30}\mathrm{Si}(n,\gamma){}^{31}\mathrm{Si} \xrightarrow{\beta^-} {}^{31}\mathrm{P} + e^- + \bar{\nu}_e, \tag{5}$$

$${}^{31}\mathrm{P}(n,\gamma){}^{32}\mathrm{P} \xrightarrow{\beta^-} {}^{32}\mathrm{S} + e^- + \bar{\nu}_e. \tag{6}$$

The native trajectory was generated with AlterBBN v2.2 at $\eta_{10} = 6.122$, with the AlterBBN `failsafe` control set to 6 and the species-abundance floor set to 10^{-50} [4]. Under this configuration, inclusion of the bridge gives

$$Y_{^{23}\mathrm{Na}} = 1.0229842 \times 10^{-31}, \tag{7}$$

$$Y_{^{30}\mathrm{Si}} = 1.0741850 \times 10^{-46}. \tag{8}$$

The compact high-resolution post-silicon diagnostic gives

$$Y_{A=31} = 1.2642940 \times 10^{-47}, \tag{9}$$

$$Y_{A=32} = 7.1605139 \times 10^{-48}. \tag{10}$$

These two values come from the compact $A \leq 40$, 32-substep convergence calculation. To keep the gap-free register on one archived numerical basis, Table 5 instead reports the single $A \leq 120$, eight-substep values for every diagnostic mass from $A = 31$ through $A = 119$. The corresponding table values at $A = 31$ and $A = 32$ are $1.2608074 \times 10^{-47}$ and $7.1659590 \times 10^{-48}$, differing from the compact values by approximately 0.276% and 0.076%, respectively.

3.1 Reduced-graph equations and numerical recurrence

The archived prescribed-trajectory calculation was executed with Python 3.13.5 and `pynucastro` 2.11.0. The $A \leq 120$ graph contains 1,079 reachable species including ^{30}Si and 2,009 directed reaction edges. It retains forward (n, γ) captures and β^- decays selected directly from ReacLib. For an edge e, the time-dependent coefficient is

$$\lambda_e(t) = \begin{cases} \rho_b(t) Y_n(t) r_e[T(t)], & e \in (n, \gamma), \\ r_e[T(t)], & e \in \beta^-, \end{cases} \tag{11}$$

In Equation (11), r_e is the corresponding ReacLib evaluation: $N_A \langle \sigma v \rangle$ for a binary neutron capture and a rate in s^{-1} for a unary beta decay [6]. The abundance equation for downstream nuclide i is

$$\frac{dY_i}{dt} = \sum_{e:d(e)=i} \lambda_e Y_{s(e)} - Y_i \sum_{e:s(e)=i} \lambda_e, \tag{12}$$

with $Y_{s(e)} = Y_{^{30}\mathrm{Si}}(t)$ for an edge sourced directly by the prescribed silicon-30 trajectory.

The species are ordered by increasing (A, Z). A neutron capture increases A, while a retained β^- edge preserves A and increases Z; consequently every retained edge points forward in the ordering and the graph is acyclic. On each native trajectory interval, T, ρ_b, Y_n, $Y_{^{30}\mathrm{Si}}$, and every λ_e are linearly interpolated to the right endpoint of each substep. The recursive backward-Euler update is

$$Y_i^{m+1} = \frac{Y_i^m + \Delta t \sum_{e:d(e)=i} \lambda_e^{m+1} \widehat{Y}_{s(e)}^{m+1}}{1 + \Delta t \sum_{e:s(e)=i} \lambda_e^{m+1}}, \tag{13}$$

In Equation (13), $\widehat{Y}_{s(e)}^{m+1}$ is the interpolated prescribed silicon-30 source for a source edge and the already updated upstream abundance for every other edge. The archived gap-free register uses eight substeps per native trajectory interval; the compact $A \leq 40$ convergence calculation uses 32.

Every downstream post-silicon species is initialized to exactly zero. The continuation reads only the stored temperature, baryon-density, free-neutron, and silicon-30 trajectories, so no abundance-floor entries from other native nuclides are imported. The calculation is one-way post-processing: downstream abundances do not alter the prescribed background or source trajectories. Reverse photodisintegration and post-silicon charged-particle channels are outside this reduced graph. The reported convergence study holds the selected rates fixed and evaluates numerical refinement of this recurrence.

A zero-silicon-30 source control leaves every downstream mass exactly zero, while half-strength and double-strength source controls reproduce one half and twice the baseline output. Under fixed background trajectories, Equation (12) is linear in the prescribed silicon-30 source; these are exact implementation controls for source isolation and linearity, not independent physical validations.

3.2 Extended qualification tests

The accepted calculation was extended without changing its physical edge selection or recurrence. The qualification matrix contains:

- a direct replay of the accepted $A \leq 120$, eight-substep result;
- outward absorbing boundaries at $A = 140$ and $A = 160$;
- temporal refinements of 16, 32, and 64 substeps on the $A \leq 160$ graph;
- exact zero-, half-, and double-source controls;
- source-cutoff controls that set stored ^{30}Si values below 10^{-50}, 10^{-49}, or 10^{-48} to zero before integration.

For replay, boundary, temporal, and source-cutoff comparisons, the implemented symmetric relative difference is

$$\delta_{\mathrm{rel}}(a, b) = \begin{cases} 0, & a = b, \\ \dfrac{|a - b|}{\max(|a|, |b|, 10^{-300})}, & a \neq b. \end{cases} \tag{14}$$

Direct stored floating-point equality is counted separately from this metric.

The enlarged $A \leq 160$ graph contains 1,767 reachable species including ^{30}Si and 3,316 directed edges. The accepted $A \leq 120$ result is reproduced with a maximum δ_{rel} of 1.7494×10^{-15}. For every mass sector from $A = 31$ through $A = 119$, the $A_{\max} = 120$, 140, and 160 calculations have identical stored eight-substep sums: all three pairwise maximum relative differences are zero, with 89 of 89 mass sectors directly equal in each comparison. This identity is expected from the forward-only topology because no state at or above the absorbing boundary can feed an upstream state. It confirms replay, graph construction, indexing, and the absence of unintended back-coupling; it is not evidence that an omitted reverse or competing physical channel would be boundary-insensitive.

Object	Count	Accounting
Native register entries, $A = 1$–30	48	Includes 46 abundance-bearing entries and the two mass-five resonances
Published nodes, $A = 31$–119	1,063	All nodes listed in the D1 and D2 register rows
Complete published register	1,111	$48 + 1{,}063$ listed nuclide or resonance entries
Archived $A \leq 120$ graph species	1,079	Silicon-30 seed, 1,063 published nodes, and 15 $A = 120$ boundary nodes
Archived graph edges	2,009	Retained forward (n, γ) and β^- edges
Additional $A = 121$–160 qualification species	688	Nodes added when the graph boundary is moved from 120 to 160
Enlarged $A \leq 160$ graph species	1,767	$1{,}079 + 688$, including the silicon-30 seed
Enlarged graph edges	3,316	Retained edges in the outward-boundary qualification

Table 2: Accounting of the published register and the two computational graphs. Register entries and graph species are different totals because the register includes the native $A = 1$–30 inventory, whereas the post-silicon graphs begin from silicon-30 as their single prescribed seed and include boundary-only nodes.

Table 2 reconciles the register and graph totals. Figure 1 shows the full mass-sector population structure from the light-state core through the qualification boundary.

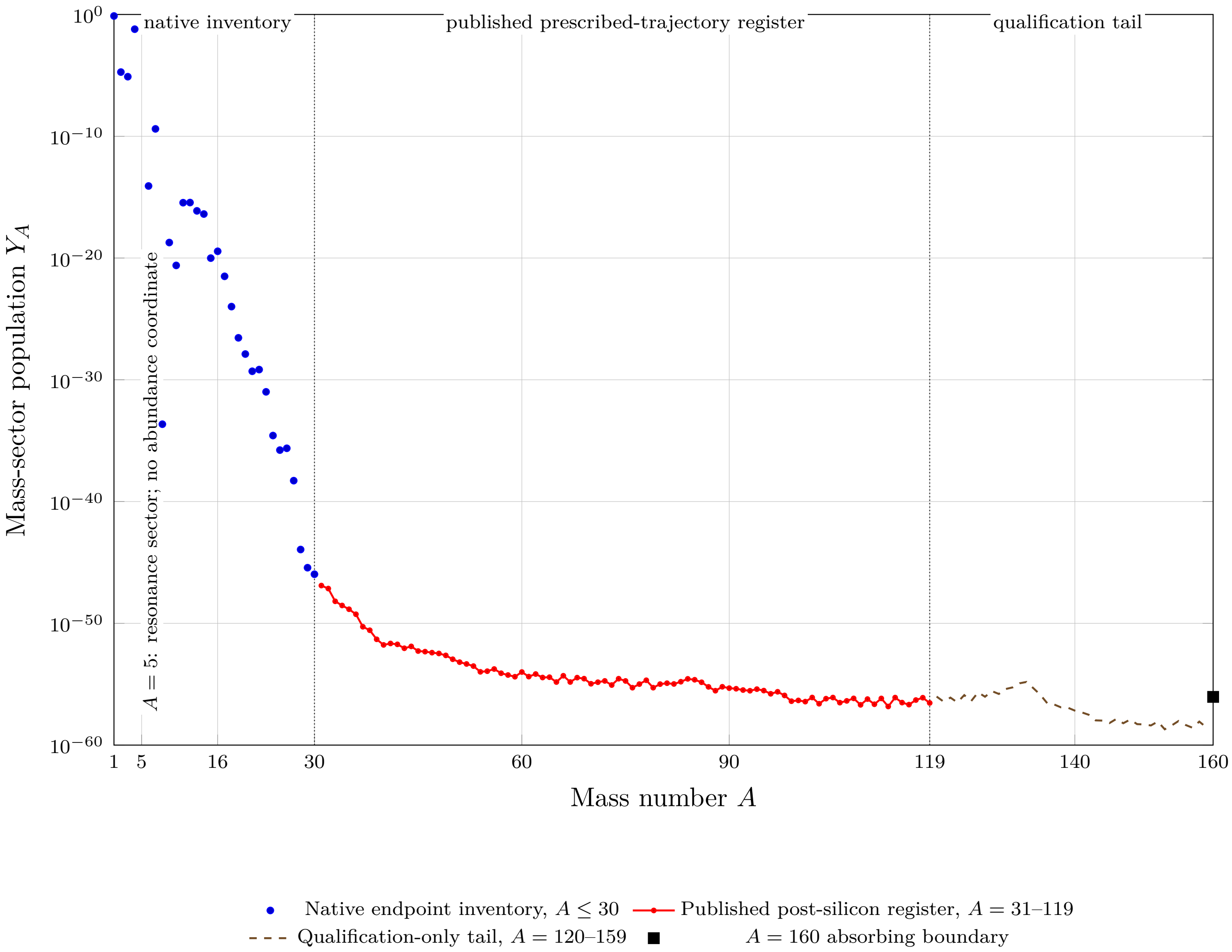

Figure 1: Mass-sector populations across the combined calculation on a logarithmic ordinate. Native endpoint sums are plotted for $A = 1$–30, with no abundance coordinate at the mass-five resonances. The archived eight-substep prescribed-trajectory sums are plotted for the published $A = 31$–119 register. The $A = 120$–159 points belong only to the enlarged $A_{\max} = 160$ boundary qualification. The final $A = 160$ marker is the absorbing-boundary population and is not assigned physical meaning. The plot displays the nearly 59-order dynamic range between the light-state core and the deepest positive qualification sectors.

A	Y_A (S8)	Y_A (S64)	S32–S64 relative change	S8–S64 relative change
31	1.260807×10^{-47}	1.264869×10^{-47}	0.0455%	0.3211%
32	7.165959×10^{-48}	7.159464×10^{-48}	0.0147%	0.0906%
40	1.687562×10^{-52}	1.567476×10^{-52}	1.0590%	7.1159%
100	4.660660×10^{-57}	4.435122×10^{-57}	0.7056%	4.8392%
119	2.873667×10^{-57}	2.740476×10^{-57}	0.6751%	4.6349%

Table 3: Selected temporal-convergence results on the enlarged $A_{\max} = 160$ graph. S8, S32, and S64 denote the number of integration substeps per stored native-trajectory interval.

Refinement	Maximum	Median	95th percentile	Below 0.1%	Below 1%	Below 5%
S8–S16	4.3172%	2.7793%	4.1617%	1/89	4/89	89/89
S16–S32	2.1940%	1.3972%	2.1088%	2/89	5/89	89/89
S32–S64	1.1059%	0.7008%	1.0613%	3/89	80/89	89/89
S8–S64	7.4514%	4.8090%	7.1784%	1/89	3/89	60/89

Table 4: Whole-register temporal-convergence distribution for the 89 published post-silicon mass sectors, $A = 31$–119. Percentages use Equation (14).

Among the selected checkpoints in Table 3, the largest S32–S64 relative change is approximately 1.06% at $A = 40$. Across all 89 published post-silicon sectors, Table 4 gives a maximum of 1.1059% at $A = 43$, a median of 0.7008%, and 80 of 89 sectors below 1%. Positive support remains present at every interior mass from $A = 31$ through $A = 159$ in the 64-substep enlarged calculation.

The source-cutoff controls sharpen the interpretation. Removing ^{30}Si source values below 10^{-50} changes $A = 31$ by only 6.10×10^{-9} relative and $A = 32$ by approximately 1.08×10^{-4} at eight substeps, but changes $A = 40$ by approximately 36.8% and suppresses selected sectors from $A = 56$ upward by more than 99.99%. At 64 substeps with the same cutoff, positive support still reaches every interior mass through $A = 159$, while the $A = 119$ magnitude falls to approximately 6.42×10^{-75}.

Here, positive support means a stored floating-point mass-sector population strictly greater than zero. It is the reachability criterion used by the graph qualification. The source-cutoff response shows that reachability and deep-tail magnitude are distinct numerical properties of the continuation.

Accordingly, the replay and boundary controls validate the implementation of the stated forward graph, while the temporal and source-cutoff results qualify its numerical reachability. The deep-tail magnitudes remain sensitive to the earliest and smallest values in the native ^{30}Si trajectory. Silicon-30 is therefore not a terminal node of this selected reduced graph. The calculation does not establish a finite physical endpoint, a precision-qualified heavy-element yield, or native-network coverage beyond silicon-30.

4 Gap-Free Nuclide Register from Mass 1 Through Mass 119

The register uses the mass-sector number abundance Y_A defined in Equation (3).

Table 5 contains 119 consecutive mass rows and 1,111 listed nuclide or resonance entries. At fixed mass number, entries belonging to different elements are isobars rather than isotopes; the table therefore uses the more precise term *nuclide*, consistent with IUPAC terminology [8]. The abundance-bearing entries at masses 1–4 and 6–30 reproduce the complete controlling native endpoint inventory. The ^{8}Be value in that source is the declared 10^{-50} abundance-floor entry and should not be interpreted as a persistent primordial abundance. Masses 31 through 119 reproduce every node in the one-way prescribed-trajectory neutron-capture and beta-minus graph. A listed diagnostic node records rate-library graph membership; it does not, by itself, establish experimental observation or measurable primordial production. Three listed graph nodes—^{43}Si, ^{47}Si, and ^{51}Si—have exactly zero endpoint abundance in the eight-substep output; every mass-sector sum from $A = 31$ through $A = 119$ remains positive.

The abundance column uses the native endpoint sums for $A = 1$–30 and preserves the archived single $A \leq 120$, eight-substep diagnostic output for $A = 31$–119. Those archived values are retained so that the register remains an exact reproduction object for the original release. Table 3 gives the stronger temporal qualification without silently replacing the archived basis. For D1 and D2 rows, Y_A is a diagnostic endpoint population in the reduced one-way graph and is not asserted to be a physical primordial heavy-element abundance. The column is marked not applicable at $A = 5$ because the listed entries are resonances rather than abundance coordinates. Numerical values are reproduced at the precision stored in the controlling source files so that the table remains an exact replay object. No physical significant-figure claim is assigned to the extra printed digits; numerical qualification is given only by the convergence and source-sensitivity controls. The evidence-class codes are defined immediately before the register.

Class codes: **N0**, native universal light-state core; **N1**, native oxygen-16 basis; **N2**, native post-oxygen extension through silicon-30; **D1**, first post-silicon prescribed-trajectory block with a separate higher-resolution diagnostic; **D2**, exploratory prescribed-trajectory graph; **R**, listed resonance entries without an abundance coordinate.

Table 5: Gap-free register of the nuclides and resonances represented in the combined calculations for every mass number from $A = 1$ through $A = 119$. The column N gives the number of listed entries at each mass.

A	N	Represented nuclides	Archived Y_A	Class
1	2	n, ^{1}H (p)	7.522677×10^{-1}	N0
2	1	^{2}H (D)	1.860436×10^{-5}	N0
3	2	^{3}H (T), ^{3}He	7.789007×10^{-6}	N0
4	1	^{4}He	6.191793×10^{-2}	N0
5	2	^{5}He, ^{5}Li	not applicable	R
6	1	^{6}Li	8.214136×10^{-15}	N1
7	2	^{7}Li, ^{7}Be	4.024159×10^{-10}	N1
8	3	^{8}Li, ^{8}Be, ^{8}B	2.246811×10^{-34}	N1
9	1	^{9}Be	1.86585×10^{-19}	N1
10	2	^{10}Be, ^{10}B	2.468639×10^{-21}	N1
11	2	^{11}B, ^{11}C	3.495781×10^{-16}	N1

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A	N	Represented nuclides	Archived Y_A	Class
12	3	^{12}B, ^{12}C, ^{12}N	3.582536×10^{-16}	N1
13	2	^{13}C, ^{13}N	7.450479×10^{-17}	N1
14	3	^{14}C, ^{14}N, ^{14}O	4.090225×10^{-17}	N1
15	2	^{15}N, ^{15}O	9.790313×10^{-21}	N1
16	1	^{16}O	3.57075×10^{-20}	N1
17	1	^{17}O	3.134831×10^{-22}	N2
18	2	^{18}O, ^{18}F	1.020704×10^{-24}	N2
19	1	^{19}F	2.790054×10^{-27}	N2
20	1	^{20}Ne	1.298913×10^{-28}	N2
21	1	^{21}Ne	5.006889×10^{-30}	N2
22	2	^{22}Ne, ^{22}Na	6.868353×10^{-30}	N2
23	2	^{23}Ne, ^{23}Na	1.022984×10^{-31}	N2
24	1	^{24}Mg	2.630659×10^{-35}	N2
25	1	^{25}Mg	1.678778×10^{-36}	N2
26	2	^{26}Mg, ^{26}Al	2.364879×10^{-36}	N2
27	1	^{27}Al	5.198167×10^{-39}	N2
28	1	^{28}Si	1.14832×10^{-44}	N2
29	1	^{29}Si	3.693064×10^{-46}	N2
30	1	^{30}Si	1.074185×10^{-46}	N2
31	2	^{31}Si, ^{31}P	1.260807×10^{-47}	D1
32	3	^{32}Si, ^{32}P, ^{32}S	7.165959×10^{-48}	D1
33	3	^{33}Si, ^{33}P, ^{33}S	6.418241×10^{-49}	D2
34	3	^{34}Si, ^{34}P, ^{34}S	2.919824×10^{-49}	D2
35	4	^{35}Si, ^{35}P, ^{35}S, ^{35}Cl	1.435181×10^{-49}	D2

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A	N	Represented nuclides	Archived Y_A	Class
36	5	^{36}Si, ^{36}P, ^{36}S, ^{36}Cl, ^{36}Ar	5.546139×10^{-50}	D2
37	5	^{37}Si, ^{37}P, ^{37}S, ^{37}Cl, ^{37}Ar	5.235818×10^{-51}	D2
38	5	^{38}Si, ^{38}P, ^{38}S, ^{38}Cl, ^{38}Ar	2.698372×10^{-51}	D2
39	6	^{39}Si, ^{39}P, ^{39}S, ^{39}Cl, ^{39}Ar, ^{39}K	4.871949×10^{-52}	D2
40	7	^{40}Si, ^{40}P, ^{40}S, ^{40}Cl, ^{40}Ar, ^{40}K, ^{40}Ca	1.687562×10^{-52}	D2
41	7	^{41}Si, ^{41}P, ^{41}S, ^{41}Cl, ^{41}Ar, ^{41}K, ^{41}Ca	2.217256×10^{-52}	D2
42	7	^{42}Si, ^{42}P, ^{42}S, ^{42}Cl, ^{42}Ar, ^{42}K, ^{42}Ca	1.8769×10^{-52}	D2
43	7	^{43}Si, ^{43}P, ^{43}S, ^{43}Cl, ^{43}Ar, ^{43}K, ^{43}Ca	8.938921×10^{-53}	D2
44	7	^{44}Si, ^{44}P, ^{44}S, ^{44}Cl, ^{44}Ar, ^{44}K, ^{44}Ca	1.271115×10^{-52}	D2
45	8	^{45}Si, ^{45}P, ^{45}S, ^{45}Cl, ^{45}Ar, ^{45}K, ^{45}Ca, ^{45}Sc	5.279289×10^{-53}	D2
46	9	^{46}Si, ^{46}P, ^{46}S, ^{46}Cl, ^{46}Ar, ^{46}K, ^{46}Ca, ^{46}Sc, ^{46}Ti	4.728164×10^{-53}	D2
47	9	^{47}Si, ^{47}P, ^{47}S, ^{47}Cl, ^{47}Ar, ^{47}K, ^{47}Ca, ^{47}Sc, ^{47}Ti	3.905347×10^{-53}	D2
48	9	^{48}Si, ^{48}P, ^{48}S, ^{48}Cl, ^{48}Ar, ^{48}K, ^{48}Ca, ^{48}Sc, ^{48}Ti	3.351341×10^{-53}	D2
49	9	^{49}Si, ^{49}P, ^{49}S, ^{49}Cl, ^{49}Ar, ^{49}K, ^{49}Ca, ^{49}Sc, ^{49}Ti	2.322291×10^{-53}	D2
50	9	^{50}Si, ^{50}P, ^{50}S, ^{50}Cl, ^{50}Ar, ^{50}K, ^{50}Ca, ^{50}Sc, ^{50}Ti	1.119949×10^{-53}	D2
51	10	^{51}Si, ^{51}P, ^{51}S, ^{51}Cl, ^{51}Ar, ^{51}K, ^{51}Ca, ^{51}Sc, ^{51}Ti, ^{51}V	6.471204×10^{-54}	D2
52	9	^{52}S, ^{52}Cl, ^{52}Ar, ^{52}K, ^{52}Ca, ^{52}Sc, ^{52}Ti, ^{52}V, ^{52}Cr	4.588623×10^{-54}	D2
53	9	^{53}S, ^{53}Cl, ^{53}Ar, ^{53}K, ^{53}Ca, ^{53}Sc, ^{53}Ti, ^{53}V, ^{53}Cr	3.067894×10^{-54}	D2
54	9	^{54}S, ^{54}Cl, ^{54}Ar, ^{54}K, ^{54}Ca, ^{54}Sc, ^{54}Ti, ^{54}V, ^{54}Cr	1.037553×10^{-54}	D2
55	10	^{55}S, ^{55}Cl, ^{55}Ar, ^{55}K, ^{55}Ca, ^{55}Sc, ^{55}Ti, ^{55}V, ^{55}Cr, ^{55}Mn	1.169812×10^{-54}	D2
56	11	^{56}S, ^{56}Cl, ^{56}Ar, ^{56}K, ^{56}Ca, ^{56}Sc, ^{56}Ti, ^{56}V, ^{56}Cr, ^{56}Mn, ^{56}Fe	1.765245×10^{-54}	D2
57	11	^{57}S, ^{57}Cl, ^{57}Ar, ^{57}K, ^{57}Ca, ^{57}Sc, ^{57}Ti, ^{57}V, ^{57}Cr, ^{57}Mn, ^{57}Fe	7.957791×10^{-55}	D2
58	11	^{58}S, ^{58}Cl, ^{58}Ar, ^{58}K, ^{58}Ca, ^{58}Sc, ^{58}Ti, ^{58}V, ^{58}Cr, ^{58}Mn, ^{58}Fe	5.671194×10^{-55}	D2
59	11	^{59}Cl, ^{59}Ar, ^{59}K, ^{59}Ca, ^{59}Sc, ^{59}Ti, ^{59}V, ^{59}Cr, ^{59}Mn, ^{59}Fe, ^{59}Co	4.114257×10^{-55}	D2

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A	N	Represented nuclides	Archived Y_A	Class
60	12	^{60}Cl, ^{60}Ar, ^{60}K, ^{60}Ca, ^{60}Sc, ^{60}Ti, ^{60}V, ^{60}Cr, ^{60}Mn, ^{60}Fe, ^{60}Co, ^{60}Ni	9.994849×10^{-55}	D2
61	12	^{61}Cl, ^{61}Ar, ^{61}K, ^{61}Ca, ^{61}Sc, ^{61}Ti, ^{61}V, ^{61}Cr, ^{61}Mn, ^{61}Fe, ^{61}Co, ^{61}Ni	4.205458×10^{-55}	D2
62	11	^{62}Ar, ^{62}K, ^{62}Ca, ^{62}Sc, ^{62}Ti, ^{62}V, ^{62}Cr, ^{62}Mn, ^{62}Fe, ^{62}Co, ^{62}Ni	6.831138×10^{-55}	D2
63	12	^{63}Ar, ^{63}K, ^{63}Ca, ^{63}Sc, ^{63}Ti, ^{63}V, ^{63}Cr, ^{63}Mn, ^{63}Fe, ^{63}Co, ^{63}Ni, ^{63}Cu	3.540761×10^{-55}	D2
64	13	^{64}Ar, ^{64}K, ^{64}Ca, ^{64}Sc, ^{64}Ti, ^{64}V, ^{64}Cr, ^{64}Mn, ^{64}Fe, ^{64}Co, ^{64}Ni, ^{64}Cu, ^{64}Zn	3.755918×10^{-55}	D2
65	12	^{65}K, ^{65}Ca, ^{65}Sc, ^{65}Ti, ^{65}V, ^{65}Cr, ^{65}Mn, ^{65}Fe, ^{65}Co, ^{65}Ni, ^{65}Cu, ^{65}Zn	1.504781×10^{-55}	D2
66	12	^{66}K, ^{66}Ca, ^{66}Sc, ^{66}Ti, ^{66}V, ^{66}Cr, ^{66}Mn, ^{66}Fe, ^{66}Co, ^{66}Ni, ^{66}Cu, ^{66}Zn	4.948077×10^{-55}	D2
67	12	^{67}K, ^{67}Ca, ^{67}Sc, ^{67}Ti, ^{67}V, ^{67}Cr, ^{67}Mn, ^{67}Fe, ^{67}Co, ^{67}Ni, ^{67}Cu, ^{67}Zn	1.517976×10^{-55}	D2
68	11	^{68}Ca, ^{68}Sc, ^{68}Ti, ^{68}V, ^{68}Cr, ^{68}Mn, ^{68}Fe, ^{68}Co, ^{68}Ni, ^{68}Cu, ^{68}Zn	3.464737×10^{-55}	D2
69	12	^{69}Ca, ^{69}Sc, ^{69}Ti, ^{69}V, ^{69}Cr, ^{69}Mn, ^{69}Fe, ^{69}Co, ^{69}Ni, ^{69}Cu, ^{69}Zn, ^{69}Ga	2.851959×10^{-55}	D2
70	13	^{70}Ca, ^{70}Sc, ^{70}Ti, ^{70}V, ^{70}Cr, ^{70}Mn, ^{70}Fe, ^{70}Co, ^{70}Ni, ^{70}Cu, ^{70}Zn, ^{70}Ga, ^{70}Ge	1.086518×10^{-55}	D2
71	13	^{71}Ca, ^{71}Sc, ^{71}Ti, ^{71}V, ^{71}Cr, ^{71}Mn, ^{71}Fe, ^{71}Co, ^{71}Ni, ^{71}Cu, ^{71}Zn, ^{71}Ga, ^{71}Ge	1.460832×10^{-55}	D2
72	12	^{72}Sc, ^{72}Ti, ^{72}V, ^{72}Cr, ^{72}Mn, ^{72}Fe, ^{72}Co, ^{72}Ni, ^{72}Cu, ^{72}Zn, ^{72}Ga, ^{72}Ge	1.863515×10^{-55}	D2
73	12	^{73}Sc, ^{73}Ti, ^{73}V, ^{73}Cr, ^{73}Mn, ^{73}Fe, ^{73}Co, ^{73}Ni, ^{73}Cu, ^{73}Zn, ^{73}Ga, ^{73}Ge	8.537773×10^{-56}	D2
74	12	^{74}Sc, ^{74}Ti, ^{74}V, ^{74}Cr, ^{74}Mn, ^{74}Fe, ^{74}Co, ^{74}Ni, ^{74}Cu, ^{74}Zn, ^{74}Ga, ^{74}Ge	2.819861×10^{-55}	D2
75	12	^{75}Ti, ^{75}V, ^{75}Cr, ^{75}Mn, ^{75}Fe, ^{75}Co, ^{75}Ni, ^{75}Cu, ^{75}Zn, ^{75}Ga, ^{75}Ge, ^{75}As	1.851494×10^{-55}	D2
76	13	^{76}Ti, ^{76}V, ^{76}Cr, ^{76}Mn, ^{76}Fe, ^{76}Co, ^{76}Ni, ^{76}Cu, ^{76}Zn, ^{76}Ga, ^{76}Ge, ^{76}As, ^{76}Se	5.175386×10^{-56}	D2
77	13	^{77}Ti, ^{77}V, ^{77}Cr, ^{77}Mn, ^{77}Fe, ^{77}Co, ^{77}Ni, ^{77}Cu, ^{77}Zn, ^{77}Ga, ^{77}Ge, ^{77}As, ^{77}Se	1.020116×10^{-55}	D2
78	12	^{78}V, ^{78}Cr, ^{78}Mn, ^{78}Fe, ^{78}Co, ^{78}Ni, ^{78}Cu, ^{78}Zn, ^{78}Ga, ^{78}Ge, ^{78}As, ^{78}Se	2.107071×10^{-55}	D2
79	13	^{79}V, ^{79}Cr, ^{79}Mn, ^{79}Fe, ^{79}Co, ^{79}Ni, ^{79}Cu, ^{79}Zn, ^{79}Ga, ^{79}Ge, ^{79}As, ^{79}Se, ^{79}Br	5.182884×10^{-56}	D2
80	14	^{80}V, ^{80}Cr, ^{80}Mn, ^{80}Fe, ^{80}Co, ^{80}Ni, ^{80}Cu, ^{80}Zn, ^{80}Ga, ^{80}Ge, ^{80}As, ^{80}Se, ^{80}Br, ^{80}Kr	1.016253×10^{-55}	D2
81	13	^{81}Cr, ^{81}Mn, ^{81}Fe, ^{81}Co, ^{81}Ni, ^{81}Cu, ^{81}Zn, ^{81}Ga, ^{81}Ge, ^{81}As, ^{81}Se, ^{81}Br, ^{81}Kr	1.210256×10^{-55}	D2
82	13	^{82}Cr, ^{82}Mn, ^{82}Fe, ^{82}Co, ^{82}Ni, ^{82}Cu, ^{82}Zn, ^{82}Ga, ^{82}Ge, ^{82}As, ^{82}Se, ^{82}Br, ^{82}Kr	1.046129×10^{-55}	D2

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A	N	Represented nuclides	Archived Y_A	Class
83	13	^{83}Cr, ^{83}Mn, ^{83}Fe, ^{83}Co, ^{83}Ni, ^{83}Cu, ^{83}Zn, ^{83}Ga, ^{83}Ge, ^{83}As, ^{83}Se, ^{83}Br, ^{83}Kr	1.597622×10^{-55}	D2
84	13	^{84}Cr, ^{84}Mn, ^{84}Fe, ^{84}Co, ^{84}Ni, ^{84}Cu, ^{84}Zn, ^{84}Ga, ^{84}Ge, ^{84}As, ^{84}Se, ^{84}Br, ^{84}Kr	2.743322×10^{-55}	D2
85	13	^{85}Mn, ^{85}Fe, ^{85}Co, ^{85}Ni, ^{85}Cu, ^{85}Zn, ^{85}Ga, ^{85}Ge, ^{85}As, ^{85}Se, ^{85}Br, ^{85}Kr, ^{85}Rb	2.316254×10^{-55}	D2
86	14	^{86}Mn, ^{86}Fe, ^{86}Co, ^{86}Ni, ^{86}Cu, ^{86}Zn, ^{86}Ga, ^{86}Ge, ^{86}As, ^{86}Se, ^{86}Br, ^{86}Kr, ^{86}Rb, ^{86}Sr	1.44283×10^{-55}	D2
87	14	^{87}Mn, ^{87}Fe, ^{87}Co, ^{87}Ni, ^{87}Cu, ^{87}Zn, ^{87}Ga, ^{87}Ge, ^{87}As, ^{87}Se, ^{87}Br, ^{87}Kr, ^{87}Rb, ^{87}Sr	6.053177×10^{-56}	D2
88	13	^{88}Fe, ^{88}Co, ^{88}Ni, ^{88}Cu, ^{88}Zn, ^{88}Ga, ^{88}Ge, ^{88}As, ^{88}Se, ^{88}Br, ^{88}Kr, ^{88}Rb, ^{88}Sr	2.926046×10^{-56}	D2
89	14	^{89}Fe, ^{89}Co, ^{89}Ni, ^{89}Cu, ^{89}Zn, ^{89}Ga, ^{89}Ge, ^{89}As, ^{89}Se, ^{89}Br, ^{89}Kr, ^{89}Rb, ^{89}Sr, ^{89}Y	6.1468×10^{-56}	D2
90	15	^{90}Fe, ^{90}Co, ^{90}Ni, ^{90}Cu, ^{90}Zn, ^{90}Ga, ^{90}Ge, ^{90}As, ^{90}Se, ^{90}Br, ^{90}Kr, ^{90}Rb, ^{90}Sr, ^{90}Y, ^{90}Zr	4.745019×10^{-56}	D2
91	14	^{91}Co, ^{91}Ni, ^{91}Cu, ^{91}Zn, ^{91}Ga, ^{91}Ge, ^{91}As, ^{91}Se, ^{91}Br, ^{91}Kr, ^{91}Rb, ^{91}Sr, ^{91}Y, ^{91}Zr	4.245786×10^{-56}	D2
92	14	^{92}Co, ^{92}Ni, ^{92}Cu, ^{92}Zn, ^{92}Ga, ^{92}Ge, ^{92}As, ^{92}Se, ^{92}Br, ^{92}Kr, ^{92}Rb, ^{92}Sr, ^{92}Y, ^{92}Zr	3.309035×10^{-56}	D2
93	15	^{93}Co, ^{93}Ni, ^{93}Cu, ^{93}Zn, ^{93}Ga, ^{93}Ge, ^{93}As, ^{93}Se, ^{93}Br, ^{93}Kr, ^{93}Rb, ^{93}Sr, ^{93}Y, ^{93}Zr, ^{93}Nb	2.900263×10^{-56}	D2
94	15	^{94}Ni, ^{94}Cu, ^{94}Zn, ^{94}Ga, ^{94}Ge, ^{94}As, ^{94}Se, ^{94}Br, ^{94}Kr, ^{94}Rb, ^{94}Sr, ^{94}Y, ^{94}Zr, ^{94}Nb, ^{94}Mo	3.931671×10^{-56}	D2
95	15	^{95}Ni, ^{95}Cu, ^{95}Zn, ^{95}Ga, ^{95}Ge, ^{95}As, ^{95}Se, ^{95}Br, ^{95}Kr, ^{95}Rb, ^{95}Sr, ^{95}Y, ^{95}Zr, ^{95}Nb, ^{95}Mo	2.978948×10^{-56}	D2
96	15	^{96}Ni, ^{96}Cu, ^{96}Zn, ^{96}Ga, ^{96}Ge, ^{96}As, ^{96}Se, ^{96}Br, ^{96}Kr, ^{96}Rb, ^{96}Sr, ^{96}Y, ^{96}Zr, ^{96}Nb, ^{96}Mo	1.634314×10^{-56}	D2
97	14	^{97}Cu, ^{97}Zn, ^{97}Ga, ^{97}Ge, ^{97}As, ^{97}Se, ^{97}Br, ^{97}Kr, ^{97}Rb, ^{97}Sr, ^{97}Y, ^{97}Zr, ^{97}Nb, ^{97}Mo	2.386422×10^{-56}	D2
98	14	^{98}Cu, ^{98}Zn, ^{98}Ga, ^{98}Ge, ^{98}As, ^{98}Se, ^{98}Br, ^{98}Kr, ^{98}Rb, ^{98}Sr, ^{98}Y, ^{98}Zr, ^{98}Nb, ^{98}Mo	1.19575×10^{-56}	D2
99	16	^{99}Cu, ^{99}Zn, ^{99}Ga, ^{99}Ge, ^{99}As, ^{99}Se, ^{99}Br, ^{99}Kr, ^{99}Rb, ^{99}Sr, ^{99}Y, ^{99}Zr, ^{99}Nb, ^{99}Mo, ^{99}Tc, ^{99}Ru	4.010949×10^{-57}	D2
100	16	^{100}Cu, ^{100}Zn, ^{100}Ga, ^{100}Ge, ^{100}As, ^{100}Se, ^{100}Br, ^{100}Kr, ^{100}Rb, ^{100}Sr, ^{100}Y, ^{100}Zr, ^{100}Nb, ^{100}Mo, ^{100}Tc, ^{100}Ru	4.66066×10^{-57}	D2
101	15	^{101}Zn, ^{101}Ga, ^{101}Ge, ^{101}As, ^{101}Se, ^{101}Br, ^{101}Kr, ^{101}Rb, ^{101}Sr, ^{101}Y, ^{101}Zr, ^{101}Nb, ^{101}Mo, ^{101}Tc, ^{101}Ru	3.713053×10^{-57}	D2

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A	N	Represented nuclides	Archived Y_A	Class
102	15	^{102}Zn, ^{102}Ga, ^{102}Ge, ^{102}As, ^{102}Se, ^{102}Br, ^{102}Kr, ^{102}Rb, ^{102}Sr, ^{102}Y, ^{102}Zr, ^{102}Nb, ^{102}Mo, ^{102}Tc, ^{102}Ru	8.071407×10^{-57}	D2
103	16	^{103}Zn, ^{103}Ga, ^{103}Ge, ^{103}As, ^{103}Se, ^{103}Br, ^{103}Kr, ^{103}Rb, ^{103}Sr, ^{103}Y, ^{103}Zr, ^{103}Nb, ^{103}Mo, ^{103}Tc, ^{103}Ru, ^{103}Rh	2.487525×10^{-57}	D2
104	16	^{104}Ga, ^{104}Ge, ^{104}As, ^{104}Se, ^{104}Br, ^{104}Kr, ^{104}Rb, ^{104}Sr, ^{104}Y, ^{104}Zr, ^{104}Nb, ^{104}Mo, ^{104}Tc, ^{104}Ru, ^{104}Rh, ^{104}Pd	6.492919×10^{-57}	D2
105	16	^{105}Ga, ^{105}Ge, ^{105}As, ^{105}Se, ^{105}Br, ^{105}Kr, ^{105}Rb, ^{105}Sr, ^{105}Y, ^{105}Zr, ^{105}Nb, ^{105}Mo, ^{105}Tc, ^{105}Ru, ^{105}Rh, ^{105}Pd	8.021328×10^{-57}	D2
106	16	^{106}Ga, ^{106}Ge, ^{106}As, ^{106}Se, ^{106}Br, ^{106}Kr, ^{106}Rb, ^{106}Sr, ^{106}Y, ^{106}Zr, ^{106}Nb, ^{106}Mo, ^{106}Tc, ^{106}Ru, ^{106}Rh, ^{106}Pd	3.052984×10^{-57}	D2
107	16	^{107}Ge, ^{107}As, ^{107}Se, ^{107}Br, ^{107}Kr, ^{107}Rb, ^{107}Sr, ^{107}Y, ^{107}Zr, ^{107}Nb, ^{107}Mo, ^{107}Tc, ^{107}Ru, ^{107}Rh, ^{107}Pd, ^{107}Ag	4.27056×10^{-57}	D2
108	17	^{108}Ge, ^{108}As, ^{108}Se, ^{108}Br, ^{108}Kr, ^{108}Rb, ^{108}Sr, ^{108}Y, ^{108}Zr, ^{108}Nb, ^{108}Mo, ^{108}Tc, ^{108}Ru, ^{108}Rh, ^{108}Pd, ^{108}Ag, ^{108}Cd	6.845493×10^{-57}	D2
109	17	^{109}Ge, ^{109}As, ^{109}Se, ^{109}Br, ^{109}Kr, ^{109}Rb, ^{109}Sr, ^{109}Y, ^{109}Zr, ^{109}Nb, ^{109}Mo, ^{109}Tc, ^{109}Ru, ^{109}Rh, ^{109}Pd, ^{109}Ag, ^{109}Cd	2.022383×10^{-57}	D2
110	16	^{110}As, ^{110}Se, ^{110}Br, ^{110}Kr, ^{110}Rb, ^{110}Sr, ^{110}Y, ^{110}Zr, ^{110}Nb, ^{110}Mo, ^{110}Tc, ^{110}Ru, ^{110}Rh, ^{110}Pd, ^{110}Ag, ^{110}Cd	5.998031×10^{-57}	D2
111	16	^{111}As, ^{111}Se, ^{111}Br, ^{111}Kr, ^{111}Rb, ^{111}Sr, ^{111}Y, ^{111}Zr, ^{111}Nb, ^{111}Mo, ^{111}Tc, ^{111}Ru, ^{111}Rh, ^{111}Pd, ^{111}Ag, ^{111}Cd	2.259657×10^{-57}	D2
112	16	^{112}As, ^{112}Se, ^{112}Br, ^{112}Kr, ^{112}Rb, ^{112}Sr, ^{112}Y, ^{112}Zr, ^{112}Nb, ^{112}Mo, ^{112}Tc, ^{112}Ru, ^{112}Rh, ^{112}Pd, ^{112}Ag, ^{112}Cd	6.772206×10^{-57}	D2
113	16	^{113}Se, ^{113}Br, ^{113}Kr, ^{113}Rb, ^{113}Sr, ^{113}Y, ^{113}Zr, ^{113}Nb, ^{113}Mo, ^{113}Tc, ^{113}Ru, ^{113}Rh, ^{113}Pd, ^{113}Ag, ^{113}Cd, ^{113}In	1.479664×10^{-57}	D2
114	17	^{114}Se, ^{114}Br, ^{114}Kr, ^{114}Rb, ^{114}Sr, ^{114}Y, ^{114}Zr, ^{114}Nb, ^{114}Mo, ^{114}Tc, ^{114}Ru, ^{114}Rh, ^{114}Pd, ^{114}Ag, ^{114}Cd, ^{114}In, ^{114}Sn	7.965147×10^{-57}	D2
115	17	^{115}Se, ^{115}Br, ^{115}Kr, ^{115}Rb, ^{115}Sr, ^{115}Y, ^{115}Zr, ^{115}Nb, ^{115}Mo, ^{115}Tc, ^{115}Ru, ^{115}Rh, ^{115}Pd, ^{115}Ag, ^{115}Cd, ^{115}In, ^{115}Sn	3.090037×10^{-57}	D2
116	17	^{116}Se, ^{116}Br, ^{116}Kr, ^{116}Rb, ^{116}Sr, ^{116}Y, ^{116}Zr, ^{116}Nb, ^{116}Mo, ^{116}Tc, ^{116}Ru, ^{116}Rh, ^{116}Pd, ^{116}Ag, ^{116}Cd, ^{116}In, ^{116}Sn	2.0995×10^{-57}	D2
117	16	^{117}Br, ^{117}Kr, ^{117}Rb, ^{117}Sr, ^{117}Y, ^{117}Zr, ^{117}Nb, ^{117}Mo, ^{117}Tc, ^{117}Ru, ^{117}Rh, ^{117}Pd, ^{117}Ag, ^{117}Cd, ^{117}In, ^{117}Sn	5.006565×10^{-57}	D2
118	16	^{118}Br, ^{118}Kr, ^{118}Rb, ^{118}Sr, ^{118}Y, ^{118}Zr, ^{118}Nb, ^{118}Mo, ^{118}Tc, ^{118}Ru, ^{118}Rh, ^{118}Pd, ^{118}Ag, ^{118}Cd, ^{118}In, ^{118}Sn	7.760345×10^{-57}	D2
119	16	^{119}Br, ^{119}Kr, ^{119}Rb, ^{119}Sr, ^{119}Y, ^{119}Zr, ^{119}Nb, ^{119}Mo, ^{119}Tc, ^{119}Ru, ^{119}Rh, ^{119}Pd, ^{119}Ag, ^{119}Cd, ^{119}In, ^{119}Sn	2.873667×10^{-57}	D2

5 The Mass-Five Gap

Mass five is included in the gap-free register as an explicit physical-gap annotation so that no integer mass row is omitted. The two contextual resonance entries,

$$^{5}\mathrm{He} \qquad \text{and} \qquad ^{5}\mathrm{Li}, \tag{15}$$

are unbound resonances:

$$^{5}\mathrm{He} \longrightarrow {}^{4}\mathrm{He} + n, \tag{16}$$
$$^{5}\mathrm{Li} \longrightarrow {}^{4}\mathrm{He} + p. \tag{17}$$

They are broad, particle-unstable resonances and do not accumulate as persistent abundance-bearing states under the conditions represented here. The row is not an exhaustive catalog of every evaluated $A = 5$ system. It records the two resonances relevant to the $n + {}^{4}\mathrm{He}$ and $p + {}^{4}\mathrm{He}$ systems and assigns neither an endpoint abundance, an independent native coordinate, nor membership in the post-silicon graph. The network can bypass the gap through channels such as

$$^{2}\mathrm{H} + {}^{4}\mathrm{He} \longrightarrow {}^{6}\mathrm{Li} + \gamma, \tag{18}$$
$$^{3}\mathrm{H} + {}^{4}\mathrm{He} \longrightarrow {}^{7}\mathrm{Li} + \gamma, \tag{19}$$
$$^{3}\mathrm{He} + {}^{4}\mathrm{He} \longrightarrow {}^{7}\mathrm{Be} + \gamma. \tag{20}$$

The mass-five row is consequently a physical gap designation, not a missing-data entry.

6 Completeness and Support Classification

The gap-free register satisfies the following checks:

$$N_{\text{mass rows}} = 119, \tag{21}$$
$$N_{\text{missing mass rows}} = 0, \tag{22}$$
$$N_{\text{listed nuclide/resonance entries}} = 1111, \tag{23}$$
$$N_{\text{zero endpoint graph nodes}} = 3, \tag{24}$$
$$A_{\min} = 1, \tag{25}$$
$$A_{\max} = 119. \tag{26}$$

The evidence classification is

$$A = 1\text{–}4,\ 6\text{–}30 : \text{native inventory}, \tag{27}$$
$$A = 5 : \text{resonance-gap annotation without an abundance coordinate}, \tag{28}$$
$$A = 31\text{–}32 : \text{first post-silicon prescribed-trajectory block, separately refined}, \tag{29}$$
$$A = 33\text{–}119 : \text{exploratory prescribed-trajectory graph}. \tag{30}$$

The nuclide lists at masses 31 through 119 describe membership in the reaction graph. They are not claims that all listed nuclides obtain measurable primordial abundances, nor are they a catalog of every experimentally known nuclide at those masses. Some ReacLib graph nodes are supported by evaluated or theoretical rates; graph membership alone is not an experimental-discovery classification. The companion nuclide-level CSV preserves the same register in machine-readable form, including A, nuclide name, proton number, endpoint abundance when applicable, evidence class, and the explicit mass-five resonance note. The three zero-endpoint graph nodes are retained because they are members of the generated reaction graph even though their final floating-point abundance is zero.

7 Scientific Boundary

The gap-free register demonstrates consecutive mass-number bookkeeping, not a continuous physical production chain. It does not erase the distinction between native-network and diagnostic support. Native-network coverage presently ends at silicon-30. The first post-silicon block is separately temporally qualified on the prescribed trajectory. The deeper tail remains an exploratory reduced-graph result.

The boundary tests show that moving the absorbing boundary from $A = 120$ to $A = 140$ and $A = 160$ leaves all 89 mass-sector sums from $A = 31$ through $A = 119$ unchanged in stored output, as required by the forward-only acyclic construction. This verifies implementation consistency but does not test the effect of omitted reverse or competing channels. The qualification also confirms that downstream states begin at exact zero and vanish identically when the ^{30}Si source is removed. Source-cutoff tests show that deep-tail magnitudes depend strongly on the early sub-10^{-50} portion of that native source trajectory. Together, the controls distinguish graph reachability from source-sensitive magnitude.

Accordingly, the current result is:

> The combined native inventory, mass-five gap annotation, and prescribed-trajectory calculation contain an explicit, gap-free archived mass register from $A = 1$ through $A = 119$. The separate $A_{\max} = 160$ qualification reproduces the published $A = 31$–119 register exactly, as required by the forward topology, and retains positive diagnostic support through its final interior mass, $A = 159$. The $A = 120$–159 qualification sectors are not presented as an extension of the published register. The result is a reduced-graph reachability and implementation test; it does not identify a finite physical endpoint or establish measurable primordial heavy-element yields.

8 Heavy-Sector Research Priorities

The detailed continuation identifies the next requirements for converting graph reachability into a physically complete heavy-sector extension. The immediate priorities are:

1. extend the native coupled network beyond silicon-30 so that the first post-silicon sectors no longer depend on a prescribed trajectory;
2. restore omitted physical channels, including reverse photodisintegration, charged-particle reactions, and alpha capture;
3. test temporal convergence across primordial and selected stellar thermodynamic histories and, after restoring reverse and competing channels, test domain-truncation sensitivity;
4. derive the minimum additional geometric coordinates needed to represent neutron excess, shell closure, beta stability, and competing transport directions without channel-by-channel retuning;

5. after constructing a physically bidirectional coupled network, enlarge the represented domain and determine whether the flow approaches a physical attenuation scale, undergoes a structural transition, or exhibits no finite endpoint within that domain.

These tasks preserve the governing principle of the universal program: one scalable geometric organization combined with reaction-specific physical rates and source directions, rather than a separate fitted geometric expression for every nucleus or mass region.

9 Conclusion

This addendum preserves the complete native, resonance-annotation, and diagnostic mass register, documents the numerical basis of the post-silicon continuation, and adds source-isolation, source-cutoff, temporal-convergence, and outward-boundary tests.

Its principal conclusion is that the tested native frontier at silicon-30 is not a terminal node of the selected reduced graph. The archived construction contains a gap-free mass-sector register from $A = 1$ through $A = 119$. The $A = 31$–119 results remain identical when the absorbing boundary is moved from $A = 120$ to $A = 140$ and $A = 160$, as required by the forward-only topology and confirmed by the implementation checks. Positive support also persists at 64-substep resolution and after removing early ^{30}Si source values below 10^{-50}, although the magnitude of the deep tail is highly source-sensitive.

The enlarged $A_{\mathrm{max}} = 160$ calculation is an implementation, boundary, and convergence qualification, not a published register extension through $A = 159$. The present result establishes numerical reachability within the selected one-way neutron-capture and beta-minus graph under the prescribed trajectories. It does not establish precision-qualified heavy-element yields, a finite physical endpoint, or native-network coverage beyond silicon-30.

The seed calculation inherits parameters introduced previously in the CPTG gravitational comparison framework, but this output-layer continuation is conditioned on those inputs and does not independently validate a cross-scale connection. Any such interpretation must be established by the companion universal-theory analysis and independent physical tests.

10 Data and Code Availability

Archived manuscript evidence bundle

`Complete-Processed-Nuclear-Chain_A1-A119_ComputationalCompanion_20260721_r02_PACKAGE.zip`
SHA-256: 0247c32e7abfb5e4446eb7ba2c5a1ecc6424493c93757f2fee3dee8a9be37d7e

The archived manuscript evidence bundle contains the preceding TeX and PDF release, machine-readable nuclide register, complete temporal-refinement outputs, qualification inputs and result tables, audit record, and a root SHA-256 manifest.

Controlling release bundle

`CPTG_Post016_Si30Bridge_AbsoluteLimitDiagnostic_20260716_r01.zip`
SHA-256: 76ee1c5c637069972c231cfda721afbcae1fb8975ca6a128ac2bea5abf79410a

Extended qualification bundle

`CPTG_Post016_Si30FloorBoundaryConvergenceQualification_20260719_r01.zip`
SHA-256: 908642caa9dbac8557faba937abcb9d1e488e3cf1d7debb787db27aa3df2c60e

CPTG repository release location

`https://github.com/CLG2025/CPTG/tree/main/nuclear-reactions`

The principal machine-readable inputs, implementation files, and claim-bearing results are:

- `RESULTS/TRACE_DIAGNOSTIC/raw_endpoint.csv`
  SHA-256 `9e234e2d8f672ad77fbfa1faa26c7fe20020e62325f07eabaa2ae5fdae1272e7`
- `RESULTS/TRACE_DIAGNOSTIC/SI30_NEUTRON_BETA_A120_S8_TRI_ENDPOINT.csv`
  SHA-256 `aecc87d9a58b5e16f2977c2857ddcd4045fd56cf5172ea397b90d1808cda8180`
- `RESULTS/TRACE_DIAGNOSTIC/SI30_NEUTRON_BETA_A120_S8_TRI_BY_MASS.csv`
  SHA-256 `a96b8914c977766e981e017a0054c40b52935a6fa5ab1a2aaf04a3567a797817`
- `DATA/Complete-Processed-Nuclear-Chain_A1-A119_IsotopeRegister_20260716_r01.csv`
  SHA-256 `f1c2b8051a40bcf3c19484341b26716b5a3ece774c5d8cc9c66afc7a6ce8e8df`
- `SOURCE/si30_neutron_beta_triangular_fast.py`
  SHA-256 `9adb230db299020e7be09fc4e5308a8af70814939242fd9b762e50ec0f73771c`
- `SOURCE/si30_floor_boundary_convergence_qualification.py`
  SHA-256 `99cb37898e65698c482c90feaea64cfa3928c95fa1a6cafb9b40b96ee694c89b`
- `SOURCE/full_temporal.py`
  SHA-256 `53224ba64ac70acc8a65a0bbe255a15b40cc86718ae2a7344b19bc284443cb81`
- `INPUT/flux_trace.csv`
  SHA-256 `cb453d3e4d6782ffe81e25609b93d8913a8663c54ecadeb480cceb5e00237f72`
- `INPUT/original_A120_rate_register.csv`
  SHA-256 `601cc379d41fe7d59e5dd3d77658050116c55b433dac47ec9ead91cdbdce6b0c`
- `RESULTS/BOUNDARY_INVARIANCE.csv`
  SHA-256 `41cbe54eebd33a8d22c6b576c56beb182c95c4225c78ab8c524989dc3acca654`
- `RESULTS/SOURCE_ISOLATION_AND_LINEARITY.csv`
  SHA-256 `db3de3c2ce7e1fe9353b73b6740b4ec23e3e841bf9d2bd593b880f8c94119a62`
- `RESULTS/SI30_SOURCE_CUTOFF_SENSITIVITY.csv`
  SHA-256 `806c27062249053e792ba445ad62d68772c4dc4657552f865f7681d0bc229f89`
- `RESULTS/SELECTED_MASS_RESULTS.csv`
  SHA-256 `eb1a5b4c3db03be30c438c6591b273515831879806826486af2060449227e6cf`
- `RESULTS/QUALIFICATION_SUMMARY.json`
  SHA-256 `2cb3d5fcc8c2b4f163ed9f0570383ea0f678960cf57950b2fa54ab75d3233bf6`

- RESULTS/TEMPORAL_CONVERGENCE_A160_FULL.csv
  SHA-256 b9562e9b1b1de8d0bf86fe5f8313e4a9fa0e9a144961f6c6a15a257514d26436
- RESULTS/TEMPORAL_CONVERGENCE_A31_A119_FULL.csv
  SHA-256 230284492adbb01e988e6288091d844a5b9641a9bf929a40c97fcda4aa7746fb
- RESULTS/TEMPORAL_CONVERGENCE_A31_A119_FULL_SUMMARY.json
  SHA-256 0ebc9fac7cd2d2aa9109eb984664c170574ce8ff25b948e6690414c1a8ca21fa

The nuclide-register CSV preserves all 1,111 listed entries and their evidence classes. The complete temporal file records the qualification values through the enlarged boundary, while the two publication-level temporal files record the 89-sector values and distribution statistics reported in Table 4.

The extended qualification bundle also contains `MANIFEST_SHA256.csv`, `requirements.txt`, `RUN_ALL_WINDOWS.cmd`, `RUN_VERIFY_WINDOWS.cmd`, and `VERIFY_PACKAGE.py`. The archived manuscript evidence bundle exposes only its package-level release verifier, `VERIFICATION/RUN_VERIFY_RELEASE_WINDOWS.cmd`, while preserving the complete original qualification archive under `QUALIFICATION/`. The root manifest records the SHA-256 identity of every packaged manuscript, input, source, result, report, and verification file. For replay, the hashed rate register and packaged environment records are controlling rather than any mutable online ReacLib snapshot.

The upstream release identities retained in the bundle are:

- CPTG_Post016_TheoryExtension_AI_Handoff_20260716_r01.zip
  SHA-256 20cbaca5676f9d06dd1cea2be6b45c9c1ecd60dfc07b75032cdc1221b2330c4a
- CPTG_Post016_RobustFrontierConfirmation_20260716_r02.zip
  SHA-256 75a1855ebf7e3460095ace4383d9c704d161a5f5a7a01cc75f1184f6a3688662
- CPTG_Ne23_BBN_vs_ExplosiveRegime_Discrimination_20260716_r01.zip
  SHA-256 8757fa29971af6f6375978ee6731b6d04c425e0916bfafc5c6a4de512ef10b60

The native trajectory was produced with AlterBBN v2.2 at $\eta_{10} = 6.122$, `failsafe`=6, and abundance floor 10^{-50}. The archived prescribed-trajectory continuation used Python 3.13.5, `pynucastro` 2.11.0, a maximum network mass of $A = 120$, and background-interpolated recursive backward Euler. The qualification used the same Python and `pynucastro` versions, extended the graph to $A = 160$, and applied the identical recurrence with Numba 0.65.1 acceleration. The qualification source contains the public nuclear-data inputs, stored native trajectories, and numerical recurrence required to reproduce the reported output-layer tests.

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